

✦ PRODUCTION HACKATHON PROTOTYPE SHOWCASE

ReguLens.AI

"Turn regulations into actionable controls."

An Agentic AI-Powered Banking Regulatory Intelligence, Traceability, and Control Assurance Platform

COGNITIVE ARCHITECTURE
11 Autonomous Agents

TRACEABILITY GUARANTEE
Unbroken End-to-End

PRODUCTION READY
Live on GitHub Pages

The Banking Regulatory Compliance Crisis

Banks navigate hundreds of circulars annually across fractured spreadsheets, inviting severe penalties.



Regulatory Velocity & Volume

Central regulators (RBI, Basel Committee, FATF, DPDP, DORA) issue hundreds of amendments annually. Critical clauses buried inside 80-page PDFs are missed or improperly classified.



Disconnected Control Frameworks

Internal policies, SOPs, and IT controls remain isolated in department silos. Changes in regulatory cadences (e.g. Annual to Quarterly) are not reflected in automated core banking schedulers.



Zero Audit Traceability & Penalties

Auditors cannot verify why a specific control was deemed sufficient for a regulatory requirement. Failure to evidence compliance exposes banks to severe statutory fines under Banking Regulation Act Section 47A.

THE REGULENS AI SOLUTION

An agentic AI nervous system that continuously reads regulations, extracts obligations, maps controls, audits evidence, and synthesizes remediation.

100% Traceable

The Unbroken Traceability Chain

Every AI finding connects source legal text to internal controls, evidence test results, and CCO approvals.



LIVE PROVENANCE CHAIN EXAMPLE (VERIFIED IN PROTOTYPE):
REG-RBI-KYC-2026 (Section 12.4) → OBL-001 (Quarterly Review) → AML-C-023 (Annual Cadence Mismatch!) → EVD-0092 (Insufficient Annual Log) → GAP-1042 (Score: 92 Critical) → REM-001 (Approved by CCO)

11 Specialized Autonomous AI Agents

A coordinated agent swarm with deterministic reasoning, confidence metrics, and role separation.

AGENT 01

Regulatory Intelligence Agent

Reads legal circulars, extracts metadata, tracks versions, and detects word-level diffs.

AGENT 02

Obligation Extraction Agent

Decomposes complex paragraphs into atomic obligations with cadence, risk, and owner.

AGENT 03

Applicability Agent

Determines scope across retail, corporate, treasury, and cross-border banking entities.

AGENT 04

Control Framework Agent

Ingests internal SOPs, controls, owners, procedures, and testing frequencies into inventory.

AGENT 05

Control Mapping Agent

Maps obligations to controls using semantic cosine matching and confidence scoring (0-100%).

AGENT 06

Evidence Assessment Agent

Audits test reports, logs, and CSV samples for sufficiency, expiration, and contradictions.

AGENT 07

Control Effectiveness Agent

Evaluates design vs operating effectiveness based on historical test executions and drift.

AGENT 08

Gap Analysis Agent

Identifies missing, weak, or cadence-mismatched controls; generates unique GAP IDs.

AGENT 09

Risk Prioritization Agent

Calculates explainable risk scores factoring regulatory severity, customer harm, and penalties.

AGENT 10

Remediation Agent

Formulates step-by-step action plans, policy adjustments, and owner milestone deadlines.

AGENT 11

Regulatory Impact Agent

Simulates the cascade blast radius of circular revisions across all downstream bank policies.

AI PROVIDER LAYER

Extensible AI Abstraction

Pluggable provider architecture supporting OpenAI, Claude, Vertex AI, or deterministic local models.

Quarterly Customer Risk Review Cadence Mismatch

A realistic end-to-end 3-minute scenario showing how ReguLens AI detects and resolves a compliance crisis.

THE REGULATORY SHOCK

RBI Master Direction - KYC (v4.2 Update)

RBI amends **Section 12.4**: Shortens customer risk review cycle from *Annual* to *Quarterly for High-Risk customers*.

- "review customer risk categorization annually for all categories."
- + "review customer risk categorization at least quarterly for High-Risk customers."

THE BANK'S VULNERABILITY

Internal Control AML-C-023

Bank control **AML-C-023** is hardcoded to run **Annually on March 31** in core banking SQL batch schedules.

Evidence EVD-0092: Evaluated as INSUFFICIENT.
Audit finding: No quarterly high-risk execution module exists in production.

GAP IDENTIFIED

GAP-1042

Score: 92/100 (Critical)

AI REMEDIATION

REM-001

Reconfigure Batch Scheduler

HUMAN IN THE LOOP

Approved

Chief Compliance Officer

AUDIT TRAIL LOGGED

Immutable

Full Traceability Preserved

Executive Dashboard & GRC Analytics

Real-time health indicators, risk distribution, and automated alerts for compliance leadership.

REGULATORY COVERAGE

87%

Assessed across 11 regulations

CONTROL COVERAGE

82%

32 internal bank controls mapped

OPEN GAPS

7

Risk-scored and triaged

CRITICAL GAPS

2

GAP-1042 & GAP-1024 (DR Backup)

Compliance Coverage by Domain

89%

AML & KYC

94%

Cybersecurity

92%

Digital Lending

98%

Treasury / LCR

78%

Data Privacy

84%

Vendor Risk

Control Effectiveness

Effective Controls

24 (75%)

Partially Effective

6 (19%)

Not Tested / Draft

2 (6%)

What-If Simulation Engine

Predict the cascade blast radius of hypothetical regulatory shifts before enforcement begins.

SCENARIO PARAMETERS

Review Cadence Contraction

Review Frequency:
Annual (365d) → Quarterly (90d)

Target Population:
High-Risk Customers (Tier 3)

Resolution SLA:
10 Working Days (Aggressive)

Traversed Cascade Dependency Graph

REGULATIONS
2

OBLIGATIONS
4

CONTROLS
7

TEAMS
4

EVIDENCE
+12

GAPS
3

Estimated Remediation Effort: ~45 person-days • Risk Exposure Increase: HIGH

Cross-Regulation Matrix & Contradiction Detection

Identify 100% control reuse and reconcile nuanced legal differences between banking circulars.

Cross-Regulation Matrix (4 Domains)

- 1. Customer Due Diligence:** RBI KYC vs FATF Rec 10/16

✓ 100% Control Reuse: AML-C-005 Sanctions Screening fulfills both.
- 2. InfoSec & Penetration Testing:** RBI Cyber vs EU DORA

✓ Harmonized: CYB-C-108 VAPT satisfies both with TIBER scope.
- 3. Data Privacy & Mobile SDK:** DPDP Act vs RBI Digital Lending

✓ Unified Guard: DAT-C-008 build-time SDK permission scanner.

Contradiction Classifier Engine

- Frequency Difference** Annual baseline vs Quarterly High-Risk requirement.
- Conditional Difference** PMLA 5-year statutory retention vs DPDP right to erasure.
- Scope Difference** RBI 6-hour technical incident report vs DPDP 72-hour privacy intimation.

Technology Stack & Production Readiness

Engineered with modern enterprise frontend technologies, strict typing, and zero runtime dependencies.

CORE ENGINE

React 18 & TypeScript

Strict mode typing across all compliance models, entities, and GRC risk factors.

BUILD & BUNDLER

Vite & Rolldown

Lightning-fast HMR and optimized production bundle compiled in < 2.0 seconds.

ENTERPRISE UI

Tailwind CSS & Recharts

Custom enterprise dark theme, responsive data tables, and dynamic SVG charts.

DEPLOYMENT

GitHub Pages (Live)

Automated deployment via gh-pages branch with relative base asset resolution.

LIVE DEMONSTRATION LINKS

Repository: <https://github.com/nihal-client33/ReguLensAI>

Live Web Application: <https://niha1-client33.github.io/ReguLensAI/>

✦ SUMMARY & TAKEAWAY

Transforming Banking Compliance with **ReguLens AI**

Moving financial institutions from disconnected manual checklists to an autonomous, traceable, and explainable control assurance ecosystem.

100% Provenance

Zero hallucination risk. Every decision maps to central bank gazette citations.

Proactive Simulation

Identify cadence bottlenecks and control gaps before regulatory audits begin.

Human in the Loop

AI recommends; Compliance Officers approve and record in immutable audit logs.